

Sub-Millijoule Intrusion Detection on a Commodity MCU Neural Processing Unit

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Abstract—We deploy an intrusion detection system (IDS) classifier on the STM32N6570-DK, a commodity ARM Cortex-M55 MCU with the Neural-ART NPU rated at 600 GOPS INT8. Exploiting the approximate equivalence between single-timestep ($T=1$) spiking neural network (SNN) inference and INT8 quantized ANN inference, we compile a lightweight MLP to the NPU without neuromorphic hardware. Evaluated on NSL-KDD (5-class) and UNSW-NB15 (10-class) with 10 random seeds, the model achieves $78.86 \pm 1.32\%$ and $64.75 \pm 0.61\%$ overall accuracy. INT8 NPU inference completes in 0.29–0.46 ms ($2.7\text{--}4.2\times$ faster than CPU-only), with an estimated energy of 44–69 μJ /inference and a Flash footprint of 120–138 KB. Compared with recent MCU-class IDS deployments on the STM32F7 (31 ms, 7.86 mJ) and Raspberry Pi 3 (27 ms), our NPU path achieves 60–100 $\times$ lower latency and over 100 $\times$ lower energy per inference. To our knowledge, this is the first IDS deployment on an ARM Cortex-M NPU and the first non-vision workload on the Neural-ART accelerator.

Index Terms—intrusion detection, spiking neural network, neural processing unit, INT8 quantization, edge AI, STM32N6

I. INTRODUCTION

Deep-learning-based intrusion detection at the IoT edge must operate under tight power and latency budgets. Recent MCU-class deployments achieve promising accuracy but at high energy cost: Chehade et al. reported 31 ms and 7.86 mJ per inference on an STM32F746G (Cortex-M7, no NPU) [1], and Diab et al. required 27 ms on a Raspberry Pi 3B+ [2]. At these latencies, CPU-only inference blocks all concurrent RTOS tasks for tens of milliseconds, making real-time packet processing difficult.

Commodity MCUs are beginning to integrate neural processing units (NPUs) that execute INT8 matrix operations at hundreds of GOPS while freeing the CPU for other tasks. A line of theoretical work [3]–[5] has shown that single-timestep ($T=1$) SNN inference is approximately equivalent to INT8 quantized ANN inference: when the membrane potential is initialized to zero, the threshold-and-fire mechanism reduces to a ReLU-like clamp. Any NPU executing INT8 `Gemm+Relu` can therefore run an SNN-equivalent model.

We put this theory to the test on the STM32N6570-DK (ARM Cortex-M55 @ 800 MHz, Neural-ART NPU, 600 GOPS INT8, \$89 dev kit). We focus on classifier inference latency, energy, and memory footprint on pre-extracted flow-level features; the full IDS pipeline (packet capture, flow aggregation) is left to future work.

Our contributions:

TABLE I
HARDWARE-DEPLOYED IDS COMPARISON. ENERGY IS PER INFERENCE.

Work	Platform	Task	Acc.	Lat.	Energy	NPU
Ngo [6]	MAX78000	bin.	98.6%	—	18 mW*	CNN eng.
Zahm [7]	Akida	multi	98.4%	—	~ 1 W*	Neurom.
Chehade [1]	STM32F7	multi	96.6%	31 ms	7.86 mJ	None
Diab [2]	RPi 3B+	multi	95.3%	27 ms	~ 6.8 mJ	None
This work	STM32N6	5/10	78.9%	0.29 ms	44 μJ	NPU

*Power, not energy; latency not reported.

- The first IDS classifier deployment on an ARM Cortex-M NPU, achieving 0.29–0.46 ms inference at an estimated 44–69 μJ .
- Empirical validation of $T=1$ SNN–ANN equivalence on commercial NPU silicon, with 99% prediction agreement between FP32 and INT8 models.
- An operator-level analysis showing that the `Floor` operator required by QCFS [3] falls back to CPU, adding 17.6% latency.

II. RELATED WORK

Table I compares IDS deployments on physical hardware. Ngo et al. [6] deployed a 1,360-parameter MLP on the MAX78000, an AI-specialized MCU with a fixed CNN accelerator, achieving 98.57% on UNSW-NB15 (binary) at 18 mW. Zahm et al. [7] used the BrainChip Akida AKD1000 neuromorphic processor (98.4%, ~ 1 W). Both use purpose-built AI/neuromorphic hardware. Chehade et al. [1] deployed a 1D-CNN on the STM32F746G (Cortex-M7, no NPU), achieving 96.59% accuracy on ISCX VPN-nonVPN but requiring 31 ms and 7.86 mJ per inference. Diab et al. [2] deployed Light-GBM/CNN on Raspberry Pi 3B+ (27–300 ms, 250–275 mW). Our work targets a general-purpose MCU with an attached programmable NPU, achieving two orders of magnitude lower energy than CPU-only MCU deployments.

Separately, several GPU-based SNN-IDS systems have been proposed [8]: convolutional SNNs [9], event-driven SNN [10], and transformer-spiking hybrids [11] report 94–99% accuracy on NSL-KDD/UNSW-NB15 but use binary classification and have no path to edge deployment. Our multi-class formulation (5/10-class) and hardware deployment target a different design point.

III. SYSTEM DESIGN

A. $T=1$ SNN-ANN Equivalence

A Leaky Integrate-and-Fire (LIF) neuron updates its membrane potential $V[t] = \beta \cdot V[t-1] + \mathbf{W}\mathbf{x}[t]$ and fires when $V[t] > \theta$. At $T=1$ with $V[0]=0$, the leak term vanishes and firing reduces to $\Theta(\mathbf{W}\mathbf{x} - \theta)$, which is equivalent to a quantized ReLU. Bu et al. [3] formalized this via QCFS; Jiang et al. [4] and Bu et al. [5] further showed that INT8 post-training quantization produces a discretization grid that closely approximates the threshold-and-fire operation. In practice, any NPU that runs INT8 Gemm+Relu is executing SNN-equivalent computation.

B. Neural-ART NPU

The STM32N6570-DK pairs an ARM Cortex-M55 at 800MHz with the Neural-ART NPU rated at 600GOPS for INT8 and 3TOPS/W efficiency [12]. The NPU supports Conv, Gemm, Relu, Add, Clip, and MaxPool in INT8; unsupported operators fall back to the Cortex-M55 CPU with automatic cache-maintenance overhead at each NPU $\leftrightarrow$ CPU handoff. The dev kit costs \$89 and includes 4.2MB internal SRAM and 128MB external NOR Flash.

C. Model Architecture

We use a four-layer MLP with BatchNorm and ReLU: $\text{Lin}(d \rightarrow 256) \rightarrow \text{BN} \rightarrow \text{ReLU} \rightarrow \text{Lin}(256 \rightarrow 256) \rightarrow \text{BN} \rightarrow \text{ReLU} \rightarrow \text{Lin}(256 \rightarrow 128) \rightarrow \text{BN} \rightarrow \text{ReLU} \rightarrow \text{Lin}(128 \rightarrow C)$, where $d=41$ (NSL-KDD) or 34 (UNSW-NB15) and $C=5$ or 10. BatchNorm is folded into the preceding Linear at ONNX export, yielding a graph of Gemm+Relu operators only (110–111K parameters). The architecture is deliberately shallow: deeper models could improve minority-class recall but risk introducing operators outside the NPU’s supported set. All hidden dimensions are powers of two (256, 128), avoiding a known Neural-ART compiler issue with prime-number channel counts [12].

D. Datasets and Training

NSL-KDD [13]: 125,973 train / 22,544 test, 41 features, 5-class. **UNSW-NB15** [14]: 175,341 train / 82,332 test, 34 features, 10-class. Training: Adam ($\text{lr}=10^{-3}$), cosine annealing, 80 epochs, inverse-frequency class weighting, 10 seeds. Three categorical features are label-encoded; remaining features are z-score normalized using training-set statistics. FP32 models are quantized to INT8 via ONNX Runtime static PTQ (MinMax, per-tensor, 1,000 calibration samples). A 24-configuration ablation (Section IV-C) confirms that the choice of calibration method, granularity, and sample size changes accuracy by at most 0.82 pp.

E. NPU Compilation and Operator Mapping

Models are compiled for the STM32N6570-DK using ST Edge AI Developer Cloud v4.0.0 (atonn v1.1.3). Table II shows the operator-level mapping. All Gemm+Relu layers map to NPU hardware epochs. The QuantizeLinear/DequantizeLinear operators at the

TABLE II
OPERATOR MAPPING TO NPU EPOCHS. ReLU INT8 ACHIEVES NEAR-100% NPU UTILIZATION; QCFS INT8 FORCES CPU FALLBACK AT EVERY FLOOR.

ONNX Operator	NPU Support	Epoch Type	Notes
Gemm (INT8)	✓	HW	Fused weight + bias
Relu	✓	HW	Fused with preceding Gemm
Clip	✓	HW	Used by QCFS
Floor	✗	SW (float)	CPU fallback, +0.08 ms
QuantizeLinear	✓	HW/Hyb	At model boundary
DequantizeLinear	✓	HW/Hyb	At model boundary

TABLE III
TEST ACCURACY (%), MEAN $\pm$ STD, 10 SEEDS). MLP IS THE ONLY MODEL ELIGIBLE FOR NPU ACCELERATION.

Dataset	Model	Overall	Macro F1	NPU?
NSL-KDD	ReLU MLP	78.86$\pm$1.32	59.20$\pm$2.80	✓
	QCFS MLP	77.82 $\pm$ 1.11	57.29 $\pm$ 2.77	Partial
	Rand. Forest	73.84 $\pm$ 0.19	47.13 $\pm$ 0.33	✗
UNSW-NB15	ReLU MLP	64.75 $\pm$ 0.61	40.29 $\pm$ 0.90	✓
	Rand. Forest	69.46$\pm$0.10	48.63$\pm$0.38	✗

model boundary may require CPU epochs depending on the input dimension.

For the ReLU INT8 path: NSL-KDD compiles to 5 HW + 1 Hyb + 2 SW epochs (0.46 ms); UNSW-NB15 compiles to 4 HW + 0 Hyb + 0 SW epochs (0.29 ms, 100% NPU). The QCFS INT8 path compiles to 13 HW + 1 Hyb + 14 SW epochs (0.54 ms) because each Floor triggers a DequantizeLinear $\rightarrow$ Floor(float) $\rightarrow$ QuantizeLinear CPU round-trip.

IV. EXPERIMENTS

A. Classification Results

Table III reports multi-seed results. On NSL-KDD, the ReLU MLP outperforms both QCFS (Wilcoxon $p=0.037$) and tree baselines. On UNSW-NB15 (10-class), Random Forest achieves higher overall accuracy, but cannot be compiled for the NPU: ST Edge AI Core rejects TreeEnsembleClassifier with “NOT IMPLEMENTED.” The accuracy gap reflects the difficulty of 10-class classification with extreme imbalance (Worms: 44 samples), not a modeling limitation specific to NPU deployment.

Per-class analysis on NSL-KDD reveals that DoS (F1=87.3%) and Normal (82.6%) are well-classified, while R2L suffers from low recall (25.0%) — 64.4% of R2L samples are misclassified as Normal. U2R (200 test samples) scatters across Probe (50.9%) and Normal (24.0%). On UNSW-NB15, Generic achieves F1=97.9%, but minority classes collapse: Worms F1=9.0%, Analysis F1=2.6%. These patterns are characteristic of extreme class imbalance, not NPU-induced degradation — FP32 and INT8 models agree on 99% of predictions.

Fig. 1 shows the NSL-KDD confusion matrix. The dominant error pattern is R2L $\rightarrow$ Normal (64.4% of R2L) and U2R $\rightarrow$ Probe (50.9% of U2R), both caused by severe class

	Predicted				
	Norm	DoS	R2L	Probe	U2R
Time	99.8%	0.0%	0.2%	0.0%	0.0%
DoS	5.2%	86.7%	0.9%	7.2%	0.0%
R2L	64.4%	0.7%	25.0%	9.5%	0.4%
Probe	3.5%	6.3%	1.0%	88.9%	0.3%
U2R	24.0%	1.5%	5.0%	50.9%	18.5%

Fig. 1. NSL-KDD confusion matrix (%). ReLU INT8, best seed. R2L→Normal (64.4%) and U2R→Probe (50.9%) dominate errors.

TABLE IV

NPU BENCHMARK ON STM32N6570-DK. ENERGY ESTIMATED FROM AN5946 [15] (~150 mW AT NOMINAL FREQUENCY). FP32 = CPU-ONLY BASELINE (0 HW EPOCHS).

Model	Dataset	Time (ms)	Energy (μ J)	HW	SW	Flash (KB)	RAM (KB)
ReLU FP32	NSL-KDD	1.24	186	0	11	466.4	2.17
ReLU INT8	NSL-KDD	0.46	69	5	2	137.7	1.25
ReLU FP32	UNSW-NB15	1.23	185	0	11	461.9	2.14
ReLU INT8	UNSW-NB15	0.29	44	4	0	120.6	0.50
QCFS INT8	NSL-KDD	0.54	81	13	14	138.0	2.00

imbalance (R2L: 2,199 train / 2,754 test; U2R: 52 train / 200 test). Normal and DoS, which dominate the training set, achieve >99% and >86% recall respectively.

B. Energy, Latency, and Memory

Table IV presents hardware measurements. INT8 NPU execution yields $2.7\times$ speedup on NSL-KDD and $4.2\times$ on UNSW-NB15 over the same model on the Cortex-M55 CPU. At an estimated 150 mW (ST AN5946 [15]), this translates to 44–69 μ J per inference — **over $100\times$** less than Chehade et al.’s 7.86 mJ on the STM32F7 [1]. UNSW-NB15 achieves 100% NPU execution (4 HW, 0 SW epochs); Flash shrinks $3.4\text{--}3.8\times$ from FP32 to INT8.

The energy figure is derived from the reference workload in AN5946, not direct measurement of our model. On-board validation with an STLINK-V3PWR probe is left to future work.

C. INT8 Quantization Robustness

A 24-configuration ablation (3 calibration methods $\times$ 2 granularities $\times$ 4 sample sizes) shows that INT8 accuracy deviates from FP32 by at most 0.82 pp across all configurations. Several INT8 configurations slightly *improve* over FP32, likely due to regularization effects of quantization noise.

Table V presents layer-wise activation comparison between FP32 and INT8 models on 1,000 NSL-KDD test samples. Hidden layers show moderate cosine similarity (0.65–0.68) due to per-neuron quantization noise, but the logit layer recovers to 0.978 and the overall prediction agreement is 99.0%. INT8 maps each activation to one of 256 discrete levels; the bounded per-neuron noise does not propagate into classification errors.

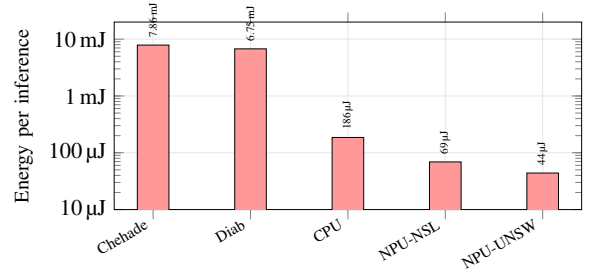


Fig. 2. Energy per inference comparison (log scale). NPU deployment achieves over $100\times$ reduction versus CPU-only MCU baselines. CPU = ReLU FP32 on Cortex-M55; NPU = ReLU INT8 on Neural-ART. Chehade [1]; STM32F7; Diab [2]; RPi 3B+.

TABLE V

LAYER-WISE FP32 VS INT8 COMPARISON (1,000 NSL-KDD SAMPLES).

Layer	Cosine Sim	MAE	L_∞
Relu_0 (256)	0.667	0.435	43.7
Relu_1 (256)	0.655	0.438	62.6
Relu_2 (128)	0.683	0.400	65.2
Logits (5)	0.978	0.103	19.8

From the $T=1$ SNN perspective, the quantized model (INT8) closely reproduces the firing decisions of the FP32 model.

V. DISCUSSION AND CONCLUSION

Design-space positioning. This work targets a different point from GPU-based IDS: sub-millisecond, sub-millijoule inference on a \$89 MCU. The absolute accuracy (78.9% NSL-KDD, 64.8% UNSW-NB15) is below GPU-simulation results, reflecting multi-class evaluation (5/10-class vs. binary) and NPU operator constraints (shallow MLP). Improving accuracy within these constraints is orthogonal to the deployment findings.

Why NPU over CPU? NPU inference frees the Cortex-M55 for RTOS tasks (packet capture, feature extraction), whereas CPU-only inference blocks the processor for 1.2 ms. Tree models achieve higher accuracy on UNSW-NB15 but have no execution path on the NPU (TreeEnsembleClassifier is not implemented in ST Edge AI Core).

QCFS barrier. The Floor operator is absent from the Neural-ART operator set, forcing CPU fallback with 17.6% latency overhead. ReLU INT8 remains the optimal activation for commodity MCU NPUs.

RTOS implications. The 0.29–0.46 ms NPU inference time fits within a 1 ms RTOS tick. Because the Cortex-M55 is idle during NPU execution, concurrent tasks (Ethernet DMA, feature extraction, display refresh) can proceed without deadline misses that would occur during the 1.2 ms CPU-only path. Validating this with a μ T-Kernel 3.0 integration is planned future work.

Limitations. We measure latency via ST Cloud benchmark; energy is estimated from the AN5946 application note, not direct measurement of our model. The evaluation covers classifier inference on pre-extracted features, not the complete

IDS pipeline. NSL-KDD is a legacy dataset, and UNSW-NB15 has extreme class imbalance that limits absolute accuracy.

Conclusion. We demonstrated sub-millisecond, sub-millijoule IDS inference on a commodity MCU NPU, achieving $2.7\text{--}4.2\times$ CPU speedup and over $100\times$ energy

reduction versus prior MCU-class deployments. Layer-wise analysis confirms 99% FP32/INT8 prediction agreement, validating $T=1$ SNN-ANN equivalence on commercial NPU silicon. All code and trained models will be publicly released.

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